

15-03-2026

Agenda:

Database-3

- SQL Aggregation
- Grouping
- Joins
- Supabase → postgres
- MongoDB Atlas

group by

department

agg

AVG

select dept, avg

IT	1
HR	2
IT	2
Fin	3
HR	1
IT	2

dept	salary
IT	1
	2
	2
HR	2
	1
Fin	3

city, count(*)

A	—
B	—
B	—
C	—
A	—

A	2
B	2
C	1

select [] from where → filter

select [] from where → filter group by []

[] having

Table created → Change schema → Alter table

DBA

INNER JOIN → matching data

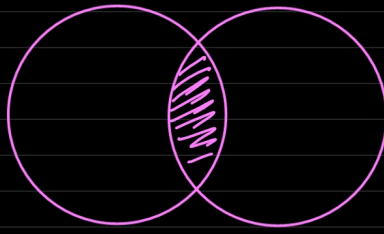
dept-id	dept-name	emp-id	name	salary	dept-id
1	A	1	A	100	1
2	B	2	B	200	2
3	C	3	C	100	2
4	D	4	D	200	A
		5	E	400	1
		6	F	500	5
		7	G	200	NULL

name, dept-name
5 row

emp-name	dept-name
A	A
B	B
C	C
D	D
E	
F	

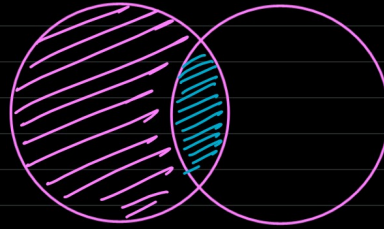
SQLite : —

Inner join

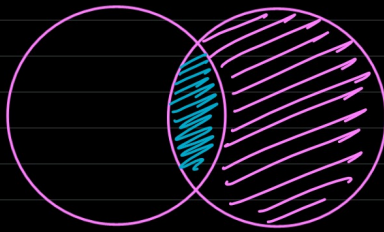


A B

left join

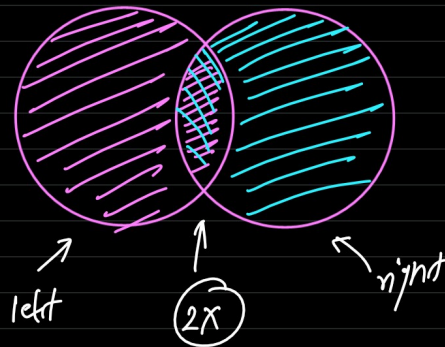


right join



full join

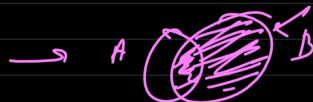
left + right - inner



(A)

left join

(B)



(B)

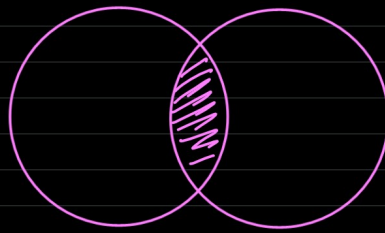
left join

A

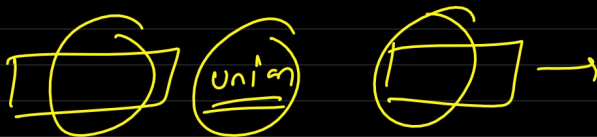
Name	dept	
A	A	inner join
B	B	
C	B	
D	C	
E	C	
F	Null	left join
Null	D	right join

Full join: left join + Right join - inner join

set → () ← append (1, 2, 3, 3, 4)



union → removes duplicates



left join union right join

Schema $\rightarrow$ Table $\rightarrow$ insert

↓
join /

id $\rightarrow$ INT
 $\swarrow$
 goto incren

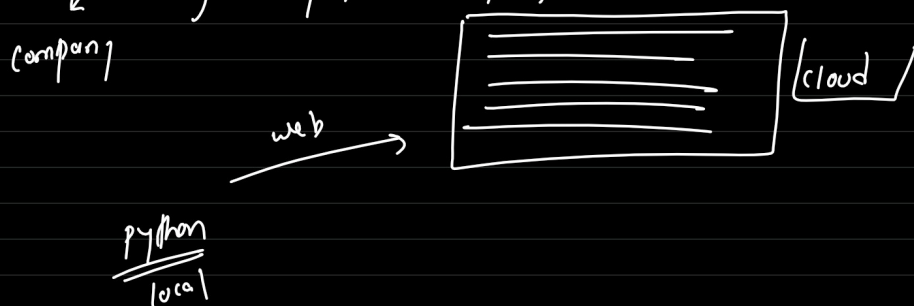
timestamp



```

graph TD
    DBMS --> PostgreSQL
    DBMS --> MongoDB
    DBMS --> MySQL
    DBMS --> Oracle
    PostgreSQL --> Relational
    MongoDB --> NonRelational
    MySQL --> NonRelational
    Oracle --> NonRelational
    
```

postgresql dbms via cloud
 | supabase |
 | login → project → postgresql DBMS instance |
 company



- int pfiles ← storage
- cv

• $p_j \leftarrow \text{file}$

PATH

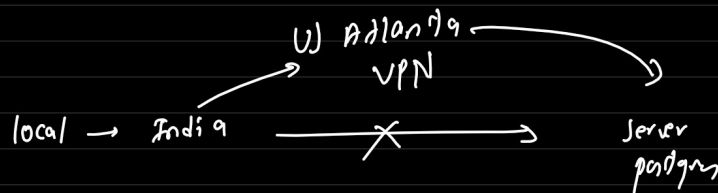
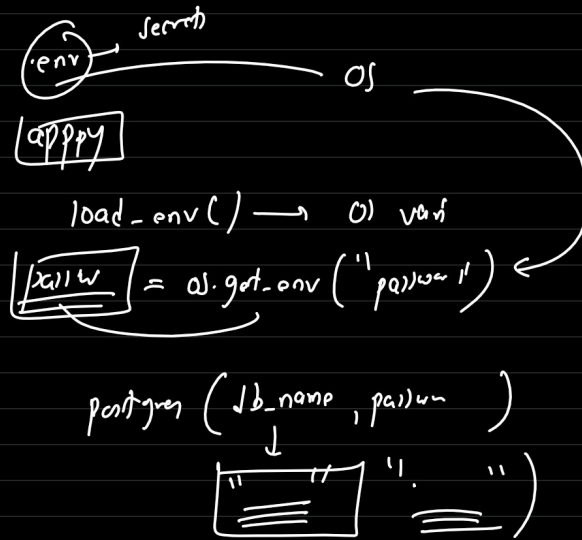
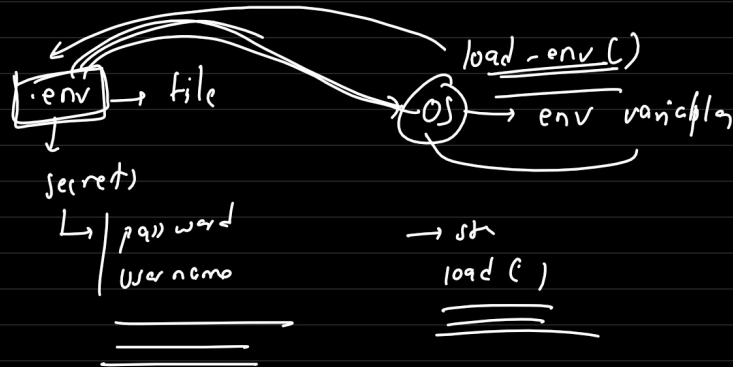
uv rot rezili



uv is not recognized

PATH ← special variable
OS

[uv,], [], []
uv.exe



Free VPN → software

windows →

Supabase → register
 → login
 → project + password
 → dashboard
 → table → schema → constraint
 → connect → python + session pooler
 ↓
 connect to postgres database
 ↓
 .env ← copy

→ project paste
 ↓
 .env
 +
 password

VPN:

→ proton VPN
 → windscribe

.py
 → import
 → load .env()
 → connection → successful → cursor
 → execute queries
 → cursor + conn → close

